
LECTURE NOTES

Discrete Mathematics notes

Sets

DM549 - Discrete Mathematics

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1 Introduction to sets

Relevant lecture handouts: Lecture 2 (slide 13), Lecture 6 (slides 5-13).

Definition (Definition 2.1.1)

A *set* (mængde) is an unordered collection of different objects, called *elements*. If object x is an element of set A , we write $x \in A$; otherwise we write $x \notin A$.

- Explicitly a set looks like $\{\text{element}_1, \text{element}_2, \dots, \text{element}_k\}$.
- A simple set $A = \{1, 2, 3, 4\}$ just means A contains the numbers 1, 2, 3, 4.
- $3 \in A$ if the object is an element of the set, $5 \notin A$ if it is not. ($\in$ and $\notin$ are written `\in` and `\notin` in LaTeX.)
- Called “set” in English, “sæt” or “mængder” in Danish.

1.1 Important sets to know

Set	Meaning
$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$	All integers.
$\mathbb{Z}^+ = \{1, 2, 3, \dots\}$	All positive integers.
$\mathbb{Z}^- = \{\dots, -3, -2, -1\}$	All negative integers.
$\mathbb{N} = \{0, 1, 2, 3, 4, \dots\}$	The natural numbers. Sometimes 0 is excluded — exercises usually state whether 0 is included.
$\mathbb{Q} = \{\frac{m}{n} \mid m \in \mathbb{Z}, n \in \mathbb{Z}^+\}$	The rational numbers: integers in fraction form. Numerator from $\mathbb{Z}$, denominator from $\mathbb{Z}^+$.
$\mathbb{R}$	The real numbers: all rationals + irrationals. Any number that can be placed on a linear number line. Contains e.g. π , $\sqrt{2}$, e . Imaginary and complex numbers are <i>not</i> included.
$\emptyset$ (<code>\emptyset</code>)	The empty set, with no elements. Also written $\{\}$.

1.2 Set-builder notation

Given a propositional / logical function, you can define a set from it where every element of the set satisfies the function’s condition.

Example: set-builder notation

Take the positive integers $\mathbb{Z}^+ = \{1, 2, 3, 4, 5, \dots\}$ and keep only the even numbers. Let $P(x)$: “ x is an even number”. Then

$$A = \{x \in \mathbb{Z}^+ \mid P(x)\} = \{2, 4, 6, 8, 10, \dots\}.$$

Intuition: include every element of $\mathbb{Z}^+$ for which the proposition $P(x)$ is true.

2 Subsets and supersets

Picture two sets A and B where A lies entirely inside B .

Definition: subset

A is a *subset* of B , written $A \subseteq B$, if ALL values in A are also in B . Correspondingly, B is a *superset* of A .

- It is typically also true that a set is a subset of itself: $A \subseteq A$, since every value in A also lies in A .
- In everyday use we often mean a *proper subset*, written $A \subset B$ (without the bar underneath).
- The point of $\subset$ versus $\subseteq$: for a proper subset we must have $A \neq B$, so a set cannot be a proper subset of itself.

3 Set operators

Set operators are the tools used to define new sets from existing ones. The examples use $A = \{1, 2, 3\}$, $B = \{2, 3, 4\}$ and the universe $U = \{1, 2, 3, 4, 5\}$, where U is every object from every set — including those not in any defined set.

3.1 Intersection

Definition: intersection

An *intersection* is the objects that two sets have in common:

$$A \cap B = \{x \mid x \in A \wedge x \in B\}.$$

Here only 2 and 3 lie in both A and B , so $A \cap B = \{2, 3\}$.

3.2 Union

Definition: union

A *union* is the collection of two sets together with the objects they share:

$$A \cup B = \{x \mid x \in A \vee x \in B\}.$$

All objects from A and B are gathered, so $A \cup B = \{1, 2, 3, 4, 5\}$.

3.3 Difference

Definition: difference

$A \setminus B$ is A with every object that also lies in B removed — only what is exclusive to A is kept. Sometimes written with $-$ (minus) instead of $\setminus$.

Here $A \setminus B = \{1\}$, because 2 and 3 are removed (they are in both A and B).

3.4 Complement

Definition: complement

The complement is everything except the set itself: take the universe and remove the set. Denoted with a bar over the set, $\overline{A}$, and found as the difference with the universe:

$$\overline{A} = U \setminus A.$$

3.5 Disjoint

Two sets are *disjoint* if their intersection is empty: $A \cap B = \emptyset$.

4 Set intervals

A set interval specifies a range the set stays within. Main takeaway: *square brackets* mean the endpoint is included, *round parentheses* mean it is not.

Definition: intervals for $a, b \in \mathbb{R}$

$[a, b] = \{x \in \mathbb{R} \mid a \leq x \leq b\}$ closed interval from a to b

$(a, b) = \{x \in \mathbb{R} \mid a < x < b\}$ open interval from a to b

$(a, b] = \{x \in \mathbb{R} \mid a < x \leq b\}$

$[a, b) = \{x \in \mathbb{R} \mid a \leq x < b\}$

Example: intervals

- $(1, 4] = \{x \in \mathbb{R} \mid 1 < x \leq 4\}$. All numbers greater than 1 up to 4: 1 is not included, but 3.12 and 4 are.
- $[3, 10) = \{x \in \mathbb{Z} \mid 3 \leq x < 10\}$. Based on $\mathbb{Z}$, so explicitly: $\{3, 4, 5, 6, 7, 8, 9\}$.

Sidenote: when an interval is based on $\mathbb{R}$ it is infinite as long as $a < b$.

5 Cardinality

Relevant lecture handouts: Lecture 8 (slides 6-10), Lecture 9 (slide 2).

5.1 Basics

For a finite set the *cardinality* is the number of elements, written $|A|$.

Example: cardinality

- $A = \{1, 3, 5, 8, 23\}$ gives $|A| = 5$.
- $\emptyset$ has cardinality 0.
- $B = \{\{1, 2, 3\}, \{4, 5, 6\}, \{7, 8, 9\}, 10, 11\}$ gives $|B| = 5$.

A set inside another set counts as only *one* object, no matter how many internal objects it has — cardinality counts only the outermost layer.

5.2 Three types of sets

Countable you can count the cardinality given enough time (e.g. A and B above).

Countably infinite you can keep going one element at a time along the number line, but never reach the end. Applies to $\mathbb{Z}, \mathbb{Z}^+, \mathbb{Z}^-, \mathbb{Q}$.

Uncountable you cannot even step to the next number. $\mathbb{R}$ is uncountable — if you are at 1, the next number is neither 1.1 nor 1.01, since you can always push the 1 further right with infinitely many 0 digits.

Definition: $\aleph_0$

Countably infinite sets have cardinality $\aleph_0$ (aleph-null) — the smallest kind of infinity. There are also $\aleph_1, \aleph_2, \aleph_3, \dots$ (larger infinities), but they are not relevant here according to the slides.

Theorem 9.30

Let A and B be finite sets. Then $|A \times B| = |A| \cdot |B|$.

6 Misc.

Relevant lecture handouts: Lecture 7 (slides 5-7).

6.1 De Morgan's laws for sets

Theorem (Example 2.2.10)

For any two sets A, B it holds that

$$\overline{A \cup B} = \overline{A} \cap \overline{B} \quad \text{and} \quad \overline{A \cap B} = \overline{A} \cup \overline{B}.$$

Remark: note the correspondence between $\cup$ and $\vee$, $\cap$ and $\wedge$, and $-$ and $\neg$.

6.2 Tuples vs. sets

Sets	Tuples
Order of objects does not matter	Order of objects matters
All objects are unique	Objects may be identical
Uses $\in$	No proper notation, so $\in$ is not used

6.3 The Cartesian product

Definition: Cartesian product

The Cartesian product pairwise combines two sets in every possible way, with the symbol $\times$:

$$A \times B = \{(a, b) \mid a \in A \wedge b \in B\}.$$

Since order matters, swapping the sets gives different objects:

Example: Cartesian product

With $A = \{1, 2\}$ and $B = \{3, 4\}$:

$$A \times B = \{(1, 3), (1, 4), (2, 3), (2, 4)\}$$

$$B \times A = \{(3, 1), (3, 2), (4, 1), (4, 2)\}$$

You can picture the product as a table. With $A = \{1, 2\}$ and $B = \{x, y\}$:

	x	y
1	$(1, x)$	$(1, y)$
2	$(2, x)$	$(2, y)$

so $A \times B = \{(1, x), (1, y), (2, x), (2, y)\}$.